

EyeVision AI — Disease classifier evaluation

Model **dis-efficientnet_b0-odir-20261006-2018** (efficientnet_b0), trained 2026-10-06T21:18:34 on **odir**.
Evaluated on: held-out TEST split (489 patients, 937 images; split by patient).

Research/educational AI estimate only. This system is not a medical diagnostic tool and does not provide a clinical diagnosis or eyeglass prescription. Please consult a qualified ophthalmologist/optometrist for diagnosis and refraction.

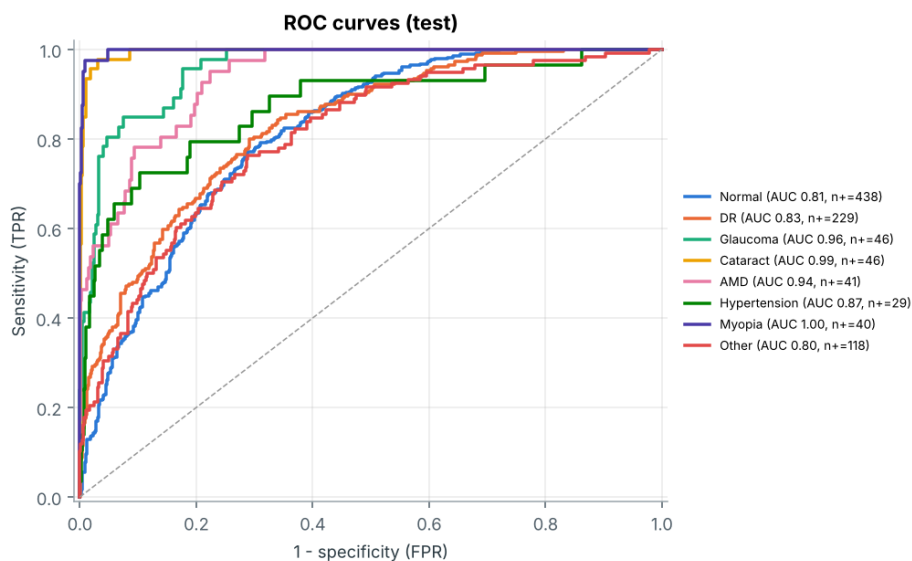
Summary (macro = mean over classes; micro = pooled over all labels)

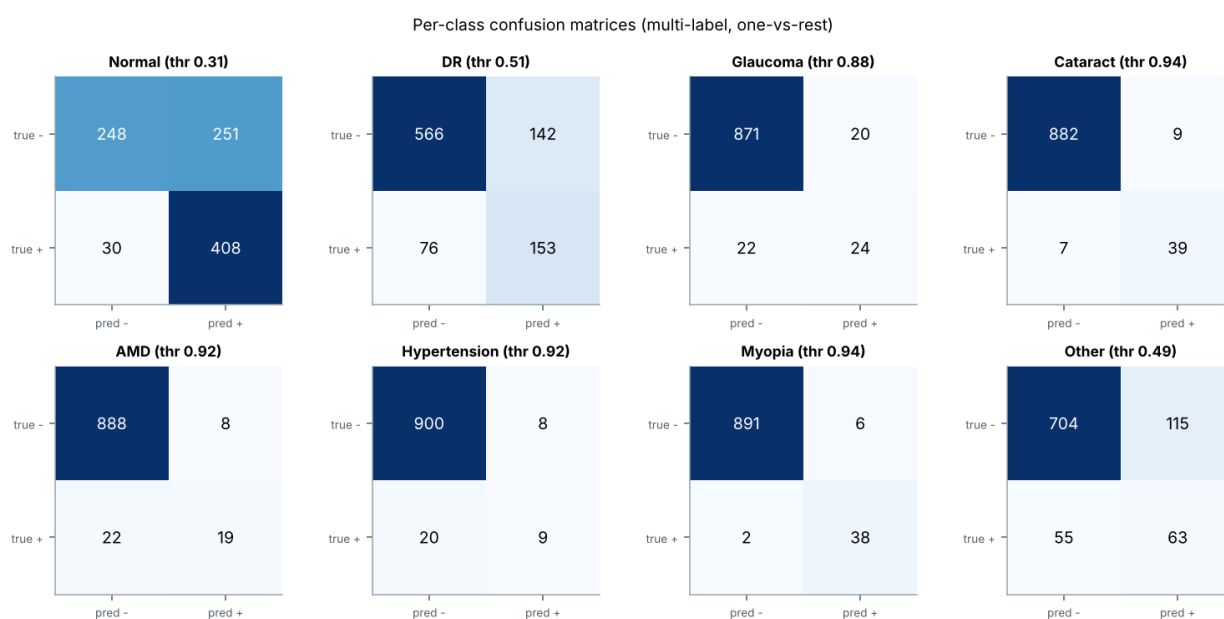
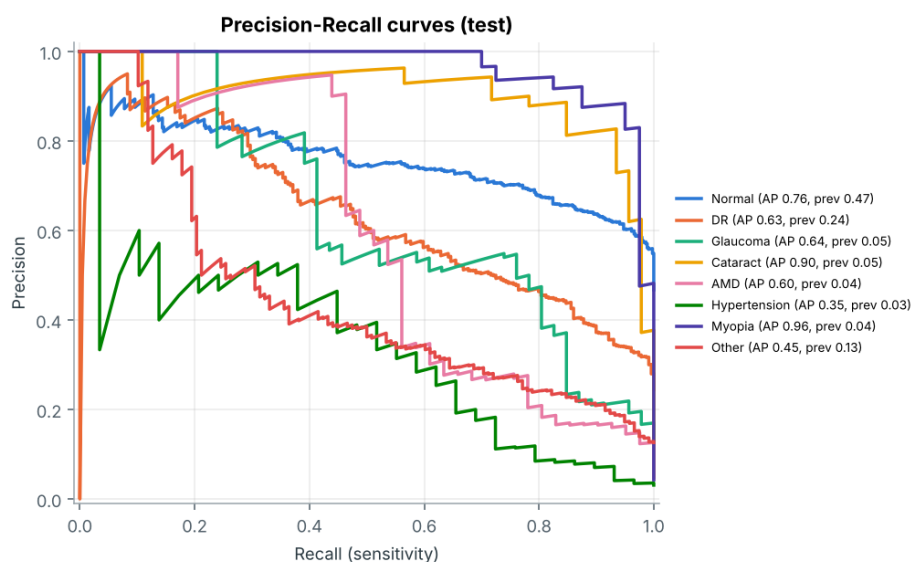
Metric	Macro	Micro
ROC-AUC	0.900	0.931
PR-AUC	0.661	0.673
F1	0.621	0.655
Precision	0.618	0.574
Sensitivity (recall)	0.653	0.763
Specificity	0.887	0.914

Per-class metrics (threshold tuned on validation set)

Class	n	n+	Thr	Prec	Sens	Spec	F1	ROC-AUC (95% CI)	PR-AUC
Normal	937	438	0.31	0.619	0.932	0.497	0.744	0.812 [0.79, 0.84]	0.758
DR	937	229	0.51	0.519	0.668	0.799	0.584	0.830 [0.80, 0.86]	0.632
Glaucoma	937	46	0.88	0.545	0.522	0.978	0.533	0.957 [0.93, 0.98]	0.636
Cataract	937	46	0.94	0.812	0.848	0.990	0.830	0.995 [0.99, 1.00]	0.902
AMD	937	41	0.92	0.704	0.463	0.991	0.559	0.935 [0.90, 0.96]	0.601
Hypertension *	937	29	0.92	0.529	0.310	0.991	0.391	0.872 [0.79, 0.94]	0.350
Myopia	937	40	0.94	0.864	0.950	0.993	0.905	0.998 [0.99, 1.00]	0.963
Other	937	118	0.49	0.354	0.534	0.860	0.426	0.803 [0.77, 0.84]	0.448

* fewer than 30 positive images — metric is unstable, interpret with caution. PR-AUC should be compared with the class prevalence (a random model's PR-AUC equals the prevalence).





Comparison with simple baselines (same test patients)

Model	Macro ROC-AUC	Macro PR-AUC	Macro F1
efficientnet_b0 (this model)	0.900	0.661	0.621
Logistic regression (features)	0.712	0.320	0.363
Random forest (features)	0.716	0.322	0.346

Subgroup analysis (rows marked * are small; differences may be noise)

variable	group	n_images	n_patients	macro_roc_auc	macro_f1	macro_sensitivity	macro_specificity
Age group	40-59	422	219	0.903	0.579	0.652	0.877
Age group	60+	457	241	0.888	0.630	0.652	0.896
Age group	<40	58	29	0.930	0.649	0.782	0.870
Sex	F	398	210	0.910	0.636	0.679	0.902
Sex	M	539	279	0.892	0.606	0.635	0.876
Eye	L	471	471	0.892	0.598	0.637	0.891
Eye	R	466	466	0.908	0.643	0.669	0.884

variable	group	n_images	n_patients	macro_roc_auc	macro_f1	macro_sensitivity	macro_specificity
Source dataset	odir	937	489	0.900	0.621	0.653	0.887